

A multiscale modeling for predicting the thermal expansion behaviors of 3D C/SiC composites considering porosity and fiber volume fraction

Zheng Sun^{a,b}, Zhongde Shan^{b,c,*}, Tianmin Shao^a, Jiahua Li^b, Xiaochuan Wu^b

^a Department of Mechanical Engineering, Tsinghua University, Beijing, 100084, China

^b State Key Laboratory of Advanced Forming Technology and Equipment, China Academy of Machinery Science and Technology, Beijing, 100044, China

^c Nanjing University of Aeronautics and Astronautics, Nanjing, 210016, Jiangsu, China

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ABSTRACT

Coefficients of thermal expansion (CTEs) are an essential design criterion of the three-dimensional carbon fiber reinforced SiC matrix composites (3D C/SiC). Representative volume element (RVE) models of microscale, void/matrix, and mesoscale developed in this work were used to investigate the CTEs of these composites. A coupled temp-displacement steady-state analysis step was created for assessing the thermal expansions behaviors of the composites by applying periodic displacement and temperature boundary conditions. Three RVE models of cuboid, hexagonal and fiber random distribution were respectively established to comparatively study the influence of fiber package pattern on the CTEs at microscale. Similarly, the effects of different void size, locations, and shapes on the CTEs of the matrix are comparatively analyzed by the void/matrix models. The prediction results at mesoscale corresponded closely to the experimental results. The effect of the porosities on the CTEs was studied by the void/matrix RVE models. The voids were effective in lowering the CTE of the 3D C/SiC composites. Furthermore, the effect of fiber volume fractions on the CTE were also taken into consideration. Equal in-plane and out-of-plane CTEs were realized by selecting appropriate fiber volume fractions for the different directions. The multiscale models developed in this work can be used to predict the thermal expansion behaviors of other complex structure composites.

1. Introduction

Carbon fiber reinforced SiC matrix composites (C/SiC) have been extensively used in the fields of aerospace, high-speed vehicles, and thermo-structure construction due to their high thermal resistance, high specific stiffness and strength, and low weight [1–5]. The properties of three-dimensional (3D) composites are superior to those of two-dimensional (2D) and two-and-a-half-dimensional (2.5D) preform structures [6,7]. However, the complex geometry structures of 3D composites, which leads to significant heterogeneity of the composites properties, represents a major drawback of these composites. In addition, the concentration of thermal stress induced by considerable temperature variations contributes significantly to fatigue and failure of 3D C/SiC composites. Investigating the thermal expansion properties of these composites is essential for avoiding potential safety hazards and improving the understanding of such composites.

Previous researchers have made greater contributions to the

investigation of the CTEs of numerous structural composites. Various theoretical models for predicting the CTEs of unidirectional (UD) composites have been proposed. Turner [8], Kerner [9,10], Schapery [11], Van to Fy [12], Chamis [13], Chamberlain [14], Geier [15], Schneider [14], Rosen & Hashin [16] have established different formulas for calculating the CTEs of UD composites based on many idealistic assumptions. However, these theoretical formulas are no longer suitable to other composites due to complex architectures and inhomogeneity of the composites.

Experimental methods and finite element analyses have attracted great attentions on the investigation of the CTEs. Through experiments, Cheng et al. [17] and Kummer et al. [18] analyzed the CTEs of C/SiC composites at different temperature points along the in-plane and out-of-plane directions. Cheng et al. [17] also determined the effect of heat treatment on the CTEs. Through finite element analysis, Ai et al. [19] and Chen et al. [20] developed two scale RVE models to evaluate the CTEs of the composites. Hu et al. [21] and Wang et al. [22]

* Corresponding author. State Key Laboratory of Advanced Forming Technology and Equipment, China Academy of Machinery Science and Technology, Beijing, 100044, China.

E-mail address: shanzd@cam.com.cn (Z. Shan).

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Table 1
Temperature-independent material parameters.

Properties	Carbon fiber	PyC interphase	SiC matrix	Voids
E_{11} (GPa)	233.13	20	460	1E-9
E_{22} (GPa)	23.1	20	460	1E-9
G_{12} (GPa)	8.97	8.13	/	/
G_{23} (GPa)	8.23	8.13	/	/
ν_{12}	0.2	0.23	0.3	1E-8
ν_{23}	0.25	0.23	0.3	1E-8
k_{11} (W/m/°C)	8	25	/	0.025
k_{22} (W/m/°C)	1			
C_p (J/g/°C)	0.7–0.9	0.14	0.14	1.01

established multi-scale models to investigate the CTEs of 3D braided composites. The effect of braiding angle, fiber volume fraction, and parameters of the carbon fiber and matrix were also discussed. Gou et al. [23] developed a general approach and discussed various boundary conditions of different representative volume element (RVE) models considering two-dimensional (2D) composites. The thermal expansion behaviors of unidirectional, plain woven, and multi-harness (4HS, 5HS, 6HS, 7HS, and 8HS) stain woven composites were systematically predicted.

Voids inevitably existed in the composites due to the imperfect and insufficient infiltration process. Kong et al. [24] established a finite element model, consisting of regular porosity, to predict the CTE of C/SiC composites. The effect of porosity on the CTE were also analyzed. Dong et al. [25,26] used the matrix-voids representative volume element models to calculate the thermo-mechanical properties of the matrix mixed voids. The cross-section of voids were treated as circle and were regularly distributed on the four corners of the matrix. In addition, Yasser et al. [27] established the Matrix-Voids models, in which the voids were assumed to be spherical, to study the effect of porosity on the CTEs of composites. Ai et al. [19] predicted the CTEs of 3D composites by the RVE models considering the void defects. They adopted a Monte Carlo algorithm to choose the elements one-by-one randomly as void defects in finite element models. Shen et al. [28] comparatively discussed the effect of naturally formed structure porous and randomly distributed defective voids on the mechanical properties of the 3D braided ceramic matrix composites.

Based on the aforementioned studies, experimental methods play a significant role in determining the CTEs of the composites, but the manufacturing cycle and production cost are time-consuming and expensive. Furthermore, assessing the thermal behaviors inside the C/SiC composites and effect of other parameters on the CTEs is impossible. The finite element analysis methods represent a promising approach for calculating the CTEs of the composites. However, few studies have focused on numerical models that predict the CTEs of 3D C/SiC composites while simultaneously taking the porosity and fiber volume

fractions into consideration.

In the present study, the CTEs associated with the in-plane and out-of-plane directions of 3D C/SiC composites are investigated via multi-scale RVE models considering the porosity and fiber volume fractions in the through-thickness direction. The local fiber volume fractions of the yarns, the porosity, the geometrical parameters of each yarn, the thickness of the interphase, and the size of the mesoscale RVE are reflected by using X-ray computed tomography. The microscale models consisted of cuboid, hexagonal and fiber random distribution, are developed for predicting the thermal and mechanical properties of the

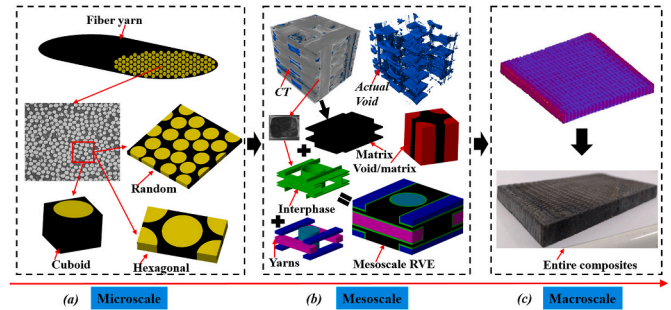


Fig. 2. Multiscale analyses procedures of 3D C/SiC composites. (a) Microscale RVE, (b) Mesoscale RVE, (c) Macroscale RVE.

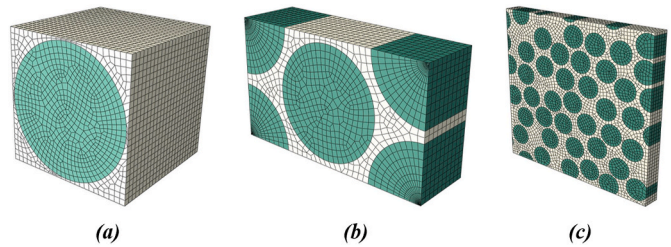


Fig. 3. Finite element models of the microscale. (a) Cuboid RVE, (b) hexagonal RVE, (c) fiber random distribution RVE.

Table 2
Geometrical parameters of the microscale RVE models.

RVE model	Size (length × width × height, μm)	Number of nodes	Number of elements
Cuboid	$8.4 \times 8.4 \times 8.4$	7416	6528
Hexagonal	$15.6 \times 9 \times 5$	9515	8430
Random	$52 \times 52 \times 5$	21,864	18,290

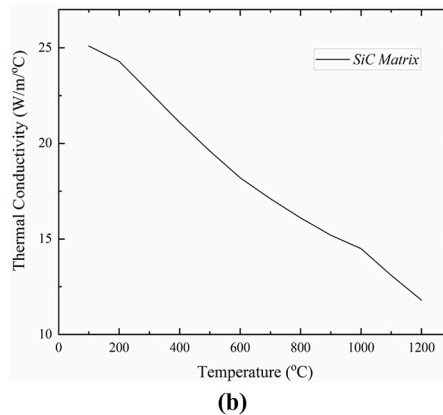
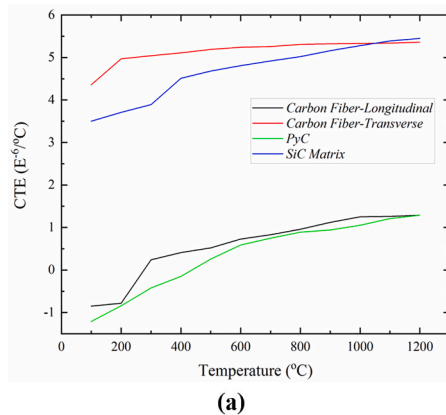


Fig. 1. Temperature-dependent variations in the materials parameters. (a) CTE of the carbon fiber, PyC and SiC matrix. (b) Thermal conductivity of the SiC matrix.

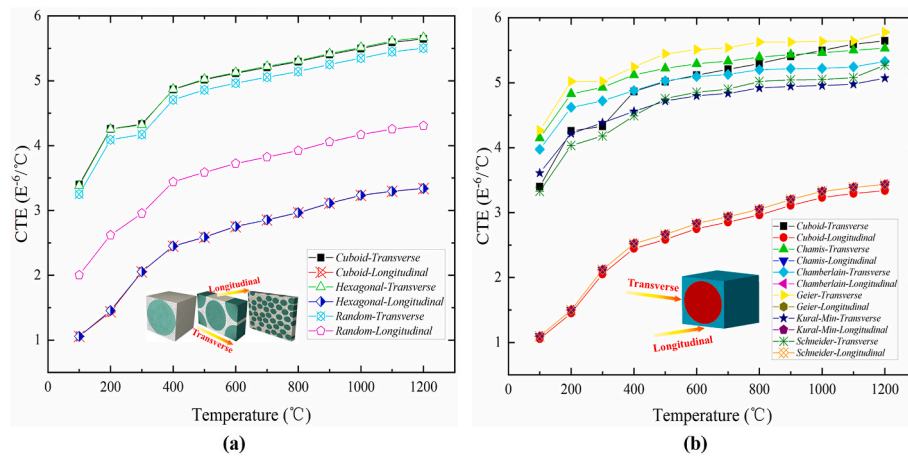


Fig. 7. The CTEs of three microscale models based on FEA and various analytical models. (a) CTEs of cuboid, hexagonal and random RVE models, (b) CTEs of cuboid and various analytical models.

structures of the composites. The model is then divided into layers to obtain the section information. The guide sleeves are fixed in the guide template, which is machined based on this information. The fiber yarns are weaved along the sleeves and the preform is thereby generated. Afterward, the sleeves are either kept by the designers or replaced in the preform. The preform is then impregnated via chemical vapor infiltration (CVI), precursor infiltration pyrolysis (PIP) or other infiltration processes.

In the present study, a 3D orthogonal structure preform based on FOWP is used to research the thermal expansion behaviors (fiber volume fraction: 46.9%). In addition, the guide sleeves are all replaced by the carbon fiber yarns. Subsequently, the preform is impregnated with a PyC interphase and SiC matrix by means of a CVI process.

Table 1 and Fig. 1 present the mechanical properties (E , G , and μ), thermal conductivity (k), and thermal expansion (α) of the carbon fiber, SiC matrix, PyC interphase, and voids [32–36]. The properties of the voids are viewed as the air [25]. The fiber is treated as a transversely isotropic material [17]. Similarly, the interphase, voids, and matrix are all treated as isotropic materials. For some of the materials, the thermal conductivity of the fiber, interphase, voids, and the mechanical properties are independent of the temperature. Therefore, these properties are considered constant, shown in Table 1. The CTEs of each constituent material and the thermal conductivity of the SiC matrix are considered temperature dependent, shown in Fig. 1.

3. Multiscale modeling

Fig. 2 illustrates the analyses procedures of the multiscale modeling for the composites. It is great significance to build the accurate models of the 3D C/SiC composites. Fortunately, the X-ray computed technology can provide the relatively accurate and high-resolution 3D models. According to the X-ray images (restructured at Sanying Precision Instruments Co., Ltd.), it is clearly found that the three-dimensional orthogonal C/SiC composites consists of fiber yarns along three directions, SiC matrix, pyrolytic carbon (PyC) interphase (white layers surrounded the yarns), cracks, and numerous voids. It is mentioned that the interphase have the same thickness and are uniformly coated along the three directions of the yarns. Additionally, two major hypotheses are proposed due to the imperfection inside the composites. The first is that the cracks in the three direction yarns are neglected due to their uncertainty and immeasurability. The second is that the voids only exist in the out-of-yarn SiC matrix.

During the multiscale simulations, the represent volume element (RVE) are usually implemented to predict the properties owing to the spatial periodicity of the composites [7]. Three RVE models of the microscale, void/matrix and mesoscale are developed in this study,

respectively. Firstly, the yarns' properties are calculated by the microscale RVE models with given fiber volume fractions. Then, the matrix properties are simulated by the void/matrix RVE models based on the actual porosities. In addition, the obtained yarns and matrix properties are transferred to the yarns and matrix in the mesoscale RVE model. At last, the mesoscale performance of the 3D orthogonal C/SiC composites compared with the experimental results are analyzed.

3.1. Microscale modeling

In order to detailedly study the properties of the microscale RVE models, the X-ray tomography technology is used to define the fiber volume fractions of the yarn. The yarns of the 3D orthogonal C/SiC composites along three directions have the same fiber volume fractions. Each yarn includes hundreds of fibers impregnated with the matrix and the package pattern is not regular. Therefore, three RVE models of cuboid, hexagonal and fiber random distribution are respectively established to investigate the influence of fiber package pattern on the thermal properties. It is mentioned that the fiber random distribution RVE model is generated by the Random Sequential Expansion method (RSE) algorithm [37]. The finite element models of three RVE models are shown in Fig. 3, respectively.

The fiber volume fractions of the yarns are consistent with the FOWP and impregnated process. Plenty X-ray figures recording the yarns' cross-section parameters are collected to determine the local fiber volume fractions. The geometry of the yarns is defined by extracting the outline of the yarns. Here the diameter of the carbon fiber is considered to be a constant $7\text{ }\mu\text{m}$. Every single fiber is assumed to be straight in each yarn. Thus the fiber volume fraction can be calculated by Eq (1).

$$F_{\text{yarn}} = \frac{n\pi r_{\text{fiber}}^2}{S} \times 100\% \quad (1)$$

where F_{yarn} stands for the local fiber volume fraction, n is the numbers of the carbon fiber in the yarn, the value of n is 3000 in each yarn. r_{fiber} represent the diameter of the carbon fiber, S stand for the area of the every yarn.

After calculating, the local fiber volume fractions are 55%. The geometrical parameters of the microscale RVE models are shown in Table 2 (see Table 3).

3.2. Void/matrix modeling

The void volume fraction is determined by the X-ray tomography technology and the value is approximately 24%. Building a precise model to reflect the actual geometrical parameters and distributions of

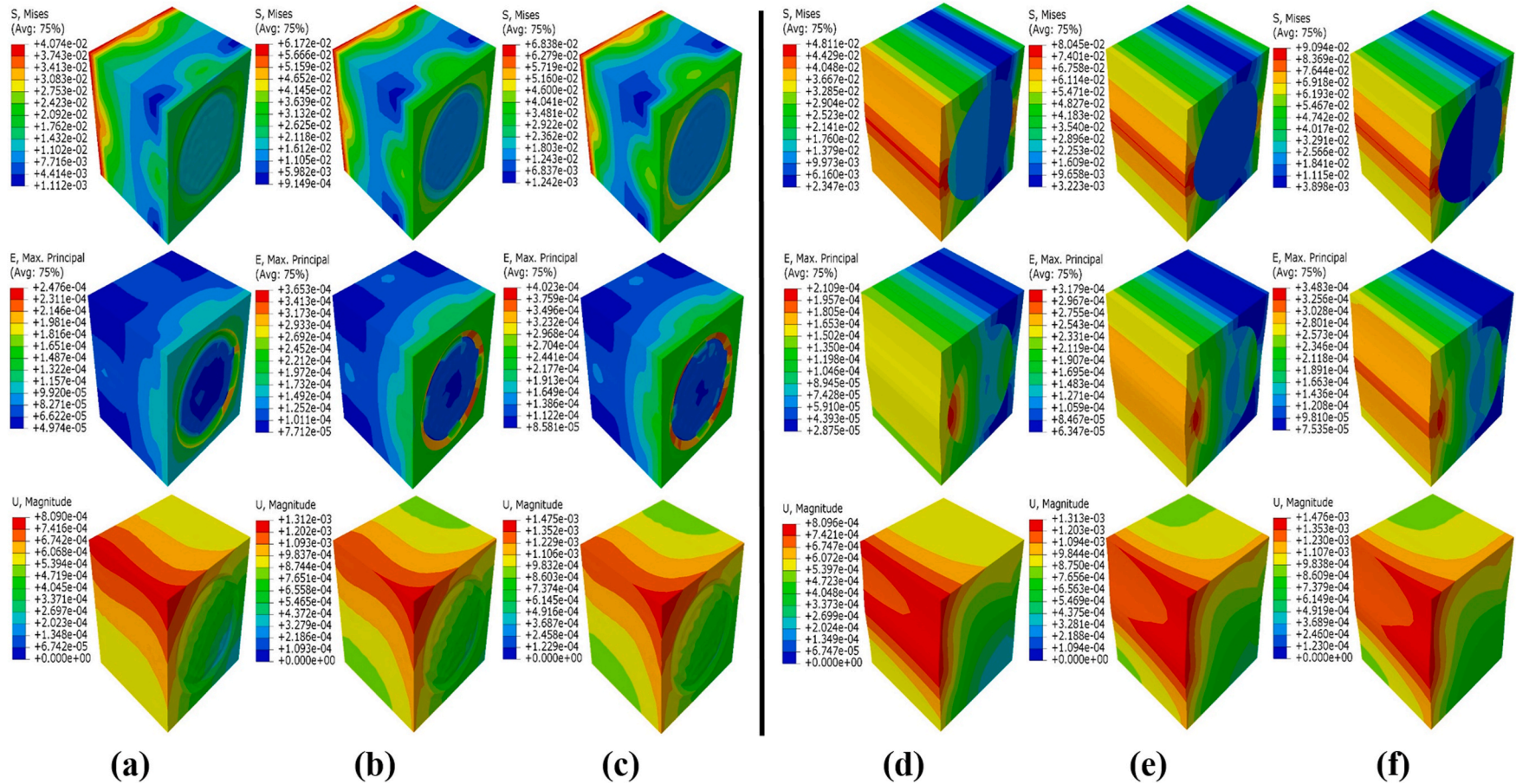


Fig. 8. Thermal mechanical contour of microscale model at different temperatures. (a)–(c): 100 °C, 600 °C, and 1200 °C, respectively, along transverse direction, (d)–(f): 100 °C, 600 °C, and 1200 °C, respectively, along longitudinal direction.

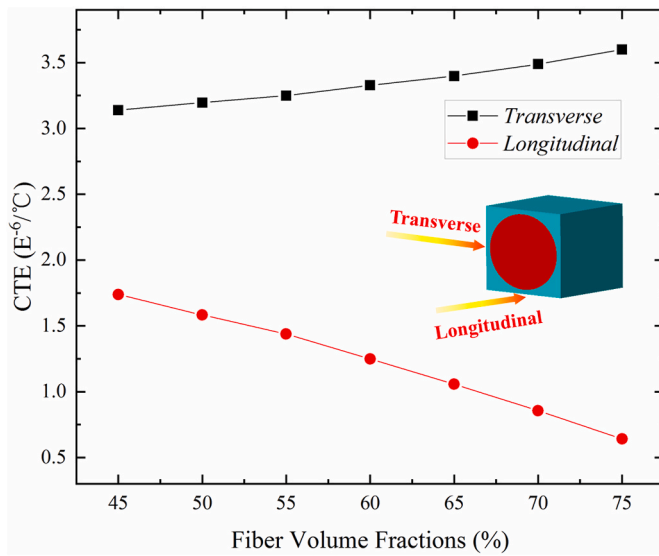


Fig. 9. CTEs of the cuboid RVE model with different fiber volume fractions.

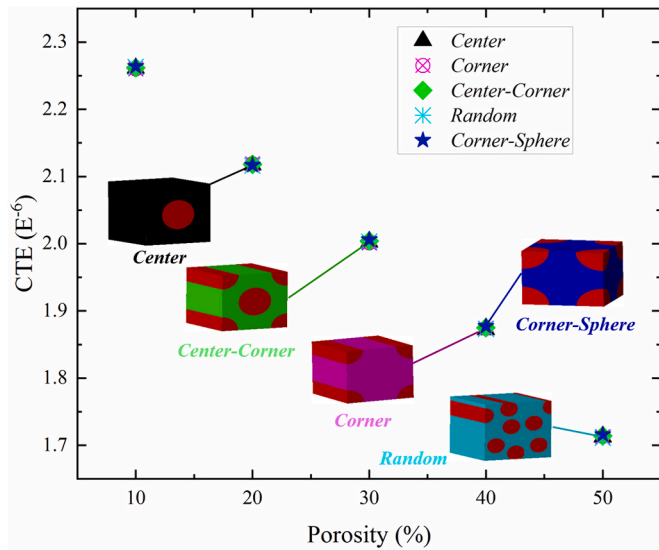


Fig. 10. Comparison of CTEs with different morphological characteristics of the voids.

the voids is difficult to realize due to their irregular shape and distributions. In this work, the void/matrix models containing of the matrix and voids are developed to calculate the properties of the matrix. In order to study the effect of different void size, locations, and shapes on the CTEs of the matrix, five models including cylinder-center cubic, cylinder-corner cubic, cylinder-center-corner cubic, cylinder random cubic, and sphere-corner cubic, are respectively established as shown in Fig. 4.

In the finite element analysis, the voids are treated as the solid air. Because the SiC matrix and the voids are all isotropic materials, the thermo-mechanical properties of the void/matrix can also be regarded as isotropic. Therefore, one direction load can enough characterize the thermo-mechanical properties of the void/matrix RVE. The sizes (length $\times$ width $\times$ height) of the void/matrix RVE models are $3 \times 3 \times 3$ mm. According to Section 40.1.1 of the Abaqus Analysis User's Manual [38], since cavity radiation effects are calculated only in heat transfer and coupled thermal-electrical procedures, the only kind of thermal-stress analysis that can include these effects is sequentially coupled thermal-stress analysis. In this study, the fully coupled

temperature-displacement analysis is adopted. Then, the effect of cavity radiation is ignored.

3.3. Mesoscale modeling

The material parameters of the yarns and SiC matrix at the mesoscale model are from the microscale and void/matrix RVE models, respectively. The geometrical parameters of the mesoscale model are developed on the basis of the X-ray tomography technology. Due to the periodicity of the C/SiC composites, the sizes of the RVE model are $2.83 \times 2.83 \times 0.98$ mm. According to the X-ray images, the cross-sections of the yarns along the x, y and z directions are considered as rectangular. The side length of the x and y yarns are 1.32 mm and 0.33 mm, and the side lengths of the z yarns are 1.17 mm and 1.07 mm. In addition, the thickness of the PyC interphase is 0.03 mm. The finite element model of the mesoscale RVE model is depicted in Fig. 5, which shows that the mesoscale model is comprised of the yarns, the PyC interphase, and the SiC matrix.

The yarns of the mesoscale model are treated as transversely isotropic entities and the material orientations are assigned via three local coordinates. Furthermore, the yarns and SiC matrix are respectively discretized. The entire structure displayed contains 224,492 elements and 78,469 nodes.

4. Finite element analyses

The RVE models of microscale and void/matrix are directly developed in the commercial software package ABAQUS/Standard (ver. 2019). After finishing the mesoscale model in software SOLIDWORKS (ver. 2015), the commercial meshing software Hypermesh (ver. 2013) was used to generate the meshes of the model. The periodic boundary conditions mentioned in subsequent sections required pairing of the corresponding nodes, and hence, the grids of the model should be spatially periodic. The finite element analyses were performed on the ABAQUS. Furthermore, the fully coupled temperature-displacement analysis was created for simulation of the thermal expansion behaviors of the composites.

4.1. Periodic boundary conditions

The periodic boundary conditions (PBC) developed by Xia and Li [39–41] are commonly used to predict the properties of a RVE model. The RVE method depends on periodic displacement boundary conditions as shown:

$$\mu_i^{j+} - \mu_i^{j-} = \bar{\epsilon}_{ik} (x_k^{j+} - x_k^{j-}) = \bar{\epsilon}_{ik} \Delta x_k^j \quad (2)$$

where $\bar{\epsilon}_{ik}$ is the average strain, x_k^{j+} is the coordinate of the model, j^+ and j^- represent the positive and negative directions, respectively, along the axis. Δx_k^j is a constant, and the $\bar{\epsilon}_{ik} \Delta x_k^j$ becomes constant with a specified strain $\bar{\epsilon}_{ik}$. This simplifies the application of PBCs in a FEA.

Fig. 6 shows the illustration of the RVE model where PBCs are applied. A is defined as a datum mark to be fixed. The points B, D and E are the master nodes. P_{x+} - P_{x-} , P_{y+} - P_{y-} , and P_{z+} - P_{z-} are pairs of the periodic nodes located at the opposite parallel boundary surfaces of the RVE model. The lines linking the points are parallel to the three coordinate axes.

Based on the periodic displacement boundary conditions described above, the periodic temperature boundary conditions can be obtained, also according to the master and slave nodes technique. The details of the periodic displacement boundary and periodic temperature boundary conditions can be found from ref 37 and 38. In this paper, we have not explain and described the specific formulas. In the finite element analysis, the implementation of periodic boundary conditions can be established by defining the linear multi-point constraint equations in

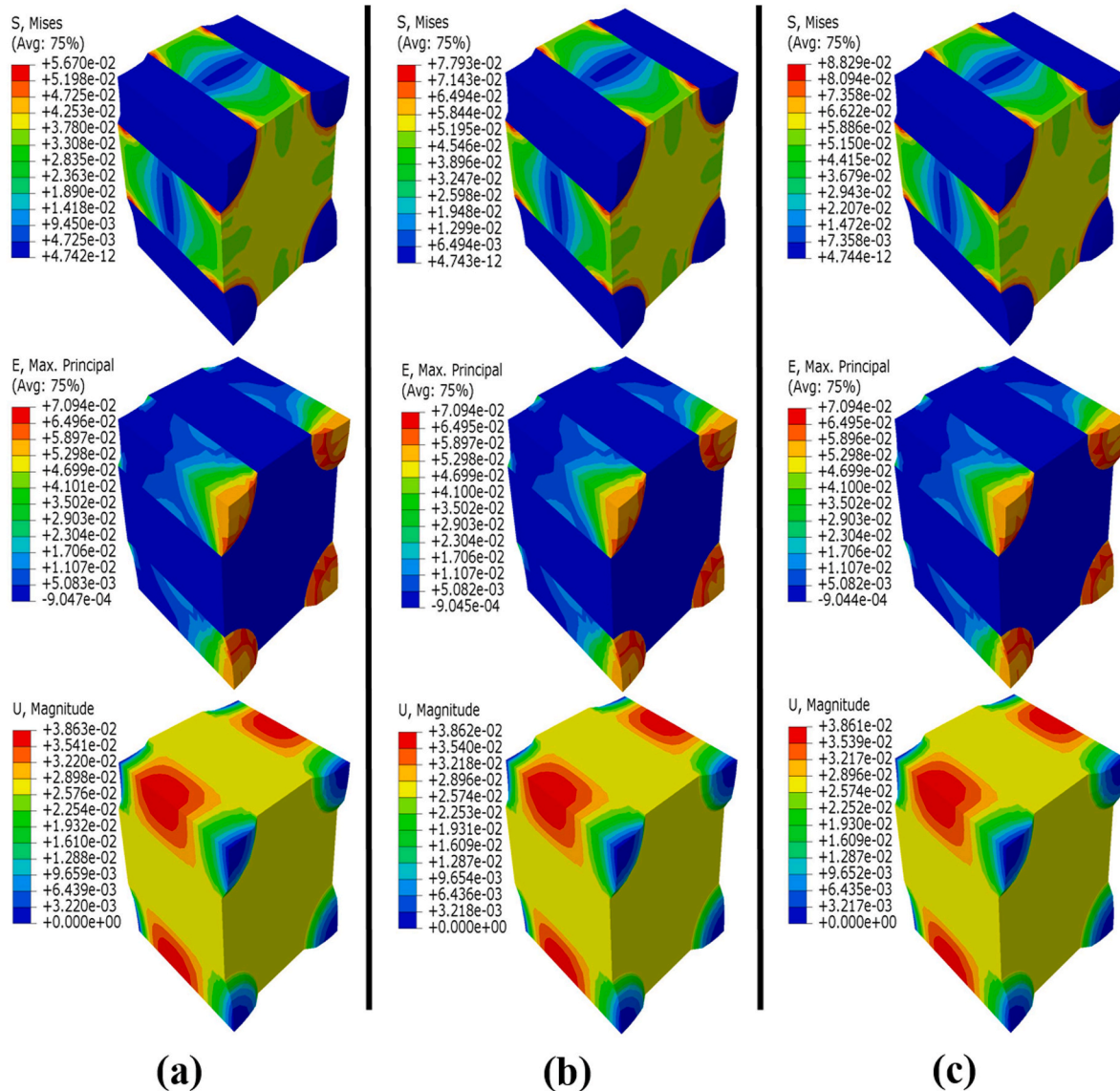


Fig. 11. Thermal mechanical contour of void/matrix model at different temperatures: (a) 100 °C, (b) 500 °C, (c) 1000 °C.

ABAQUS. The meshes and the nodes on the two opposite parallel boundary surfaces should be the same. According to the ABAQUS manual [37], the displacement degrees of freedom characterizing the three directions are 1, 2, and 3, and the temperature degree of freedom is 11.

4.2. Calculation of mechanical properties

The mechanical properties of the microscale model and void/matrix model can be obtained from the coupled thermal-stress analysis. The periodic displacement boundary conditions will be applied to the RVE models, and the average stress and average strain can be calculated as follows:

$$\bar{\sigma}_{ij} = \frac{1}{V} \int \sigma_{ij} dv \quad (3)$$

$$\bar{\varepsilon}_{ij} = \frac{1}{V} \int \varepsilon_{ij} dv \quad (4)$$

where, $\bar{\sigma}_{ij}$ and $\bar{\varepsilon}_{ij}$ are the average stress and average strain of the RVE model, respectively. V is the volume of the RVE model. σ_{ij} and ε_{ij} is the

stress and strain of every element. i and j represent the coordinates along the x , y and z directions.

The elastic constants of the model can be obtained as follows:

$$E_i = \frac{\bar{\sigma}_i}{\bar{\varepsilon}_i}, \mu_{ij} = -\frac{\bar{\varepsilon}_j}{\bar{\varepsilon}_i}, G_{ij} = \frac{\bar{\sigma}_{ij}}{\bar{\varepsilon}_{ij}}, \quad (5)$$

where, E_i , μ_{ij} and G_{ij} are the Young modulus, Poisson's ratio and shear modulus, respectively. i and j refer to the coordinates along the x , y and z directions. By defining six independent sets of strain in the FEA, the six sets of stress can be calculated. Therefore, the mechanical properties of the microscale model and void/matrix model can be determined from Eq. (5).

4.3. Calculation of thermal conductivity

For calculating the thermal conductivity of the microscale and void/matrix model, the constitutive equations can be determined from Fourier's law:

$$q_i = k_i \Delta T_i \quad (6)$$

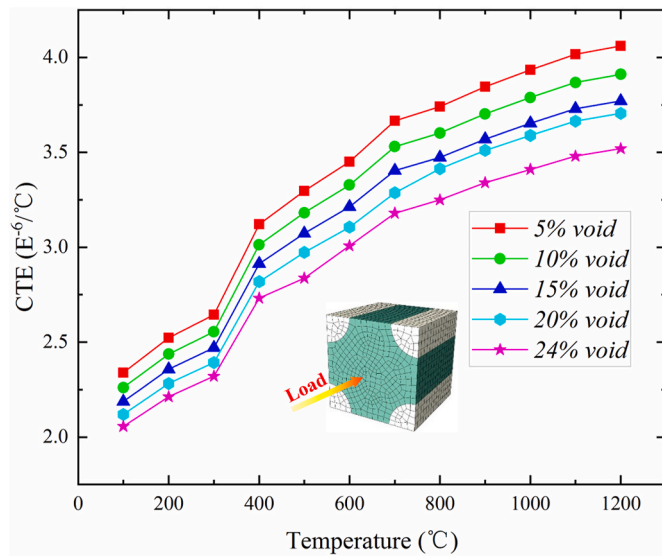


Fig. 12. CTE of void/matrix model at different temperatures and various porosities.

where, q_i, k_i and ΔT_i are the heat flux, thermal conductivity and temperature gradient, and i refers to the coordinate in x, y and z directions. The effective thermal conductivities of the RVE model can be obtained from Eq. (6).

4.4. Calculation of thermal expansion

For calculating the thermal expansion, the constitutive equations can be described by Eq. (7), which considers the temperature change. The thermal strain induced by temperature changes can be calculated from Eq. (8).

$$\sigma = C(\varepsilon - \alpha \Delta T) \quad (7)$$

$$\varepsilon_i = \alpha_i \Delta T \quad (8)$$

where $\sigma, \varepsilon, \alpha$ and ΔT are the thermal stress, stiffness matrix, thermal strain, coefficient of thermal expansion and temperature offset from the reference temperature, respectively. The reference temperature is assumed to be 0 °C in present work. The subscript i refers to coordinates in x, y and z directions. When only the temperature loads are considered, σ is equal to zero and, correspondingly, the strain ε is equal to the thermal strain $\alpha_i \Delta T$. For calculating the CTE along x direction, the temperature T is applied to the RVE model along this direction, the ε_x can then be substituted into Eq. (8) for calculation of the expansion.

5. Results and discussion

5.1. Microscale model

Here we firstly discussed the CTEs of the cuboid, hexagonal and fiber random RVE models with the same fiber volume fractions. The CTEs of three models at different temperature points are illustrated in Fig. 7 (a). As shown in the figure, the values are increasing with the increment of the temperature. The transverse thermal expansions are always higher than the longitudinal CTEs. The results of the cuboid and hexagonal RVE models are almost same along both the transverse and longitudinal directions. However, the transverse CTEs of the fiber random distribution RVE is slightly lower than the CTEs of the cuboid and hexagonal while the longitudinal CTEs are substantially larger than the other RVE models. In order to verify the accuracy and reliability, the values of the cuboid RVE model is compared with several analytical models mentioned in Section 1. The microscale and analytical models results are shown in

Fig. 7 (b), which reveals a close correspondence between the longitudinal CTEs determined via these methods. The transverse CTEs differ only slightly. The cuboid RVE model is used to study the thermal mechanical behaviors and the effect of the fiber volume fractions on the CTEs. The thermal mechanical behaviors of the cuboid RVE model at different temperature points are illustrated in Fig. 8. From the figure, the thermal stress, thermal strain, and thermal displacement among the longitudinal and transverse directions are temperature dependent, and the value of each parameter increases with increasing temperature. Furthermore, the distributions of these parameters are uneven along both the transverse and longitudinal directions. To determine the influence of fiber volume fractions of the microscale on the CTEs, the cuboid RVE models with fiber volume fractions of 45%, 50%, 55%, 60%, 65%, 70%, and 75% are built, respectively. The CTEs related to different fiber volume fractions and temperature are shown in Fig. 9. It can be concluded that the transverse CTEs of the cuboid model slightly increases while the longitudinal CTEs substantially decreases as increasing the fiber volume fractions. Notably, the transverse CTEs increases by 14.65% when the fiber volume fractions increases from 45% to 75% at the temperature of 100 °C. Similarly, the longitudinal CTEs decreases by 63.03%. The fiber volume fractions of the microscale have greater effect on the longitudinal CTEs than the transverse ones.

5.2. Void/matrix model

According to the X-ray images, the distributions and shapes of the voids are irregular. In order to investigate the effects of void sizes, locations, and shapes on the CTEs of the void/matrix models, five models containing of different morphological characteristics of voids are respectively established in Section 3.2, including cylinder-center cubic, cylinder-corner cubic, cylinder-center-corner cubic, cylinder random cubic, and sphere-corner cubic. Each model has the same void volume fractions. Fig. 10 shows the variations of the CTEs with different voids characteristics. One can find that the results of CTEs agree well, which demonstrates that the morphological characteristics of the voids have little influence on the CTEs. Therefore, the cylinder-corner model was used for calculation of the CTE for various composites with different structures.

The thermal stress, thermal strain, and thermal displacement of the SiC matrix with 24% porosity are shown in Fig. 11. To determine the influence of porosity on the CTEs, void/matrix models with 5% porosity, 10% porosity, 15% porosity, and 20% porosity are built. The CTEs associated with different porosities and temperatures are shown in Fig. 12. As shown in Figs. 11 and 12, the CTEs and thermal strain increase with increasing temperature. The thermal stress concentration always appears on the border between the SiC matrix and the voids. Furthermore, the CTEs of the void/matrix model decreases with increasing porosity. The voids are effective in lowering the CTEs of the model. For example, the CTEs decreases by 13.33% when the porosity increases from 5% to 24% at a temperature of 1200 °C.

5.3. Mesoscale model

A DIL 402C dilatometer (NETZSCH Co., Germany) was used to measure the CTEs of $5 \times 5 \times 25$ mm test specimens, which were machined through the in-plane and out-of-plane directions of the composites. The numerical results and experimentally determined values are shown in Table 2, which reveals a close coincidence between the two sets of the values. In general, both numerical and experimental results indicate that the in-plane and out-of-plane CTEs increase slightly with increasing temperature, where they have maximum values of 2.611 (experiment: 3.024) and 2.343 (experiment: 2.516) $10^{-6}/^{\circ}\text{C}$. Additionally, for a given temperature, the in-plane CTEs are always greater than the out-of-plane CTEs. For temperatures lower than 200 °C, the RVE model for predicting the in-plane CTEs is more accurate than the model for predicting the out-of-plane CTEs, but the reverse is true for

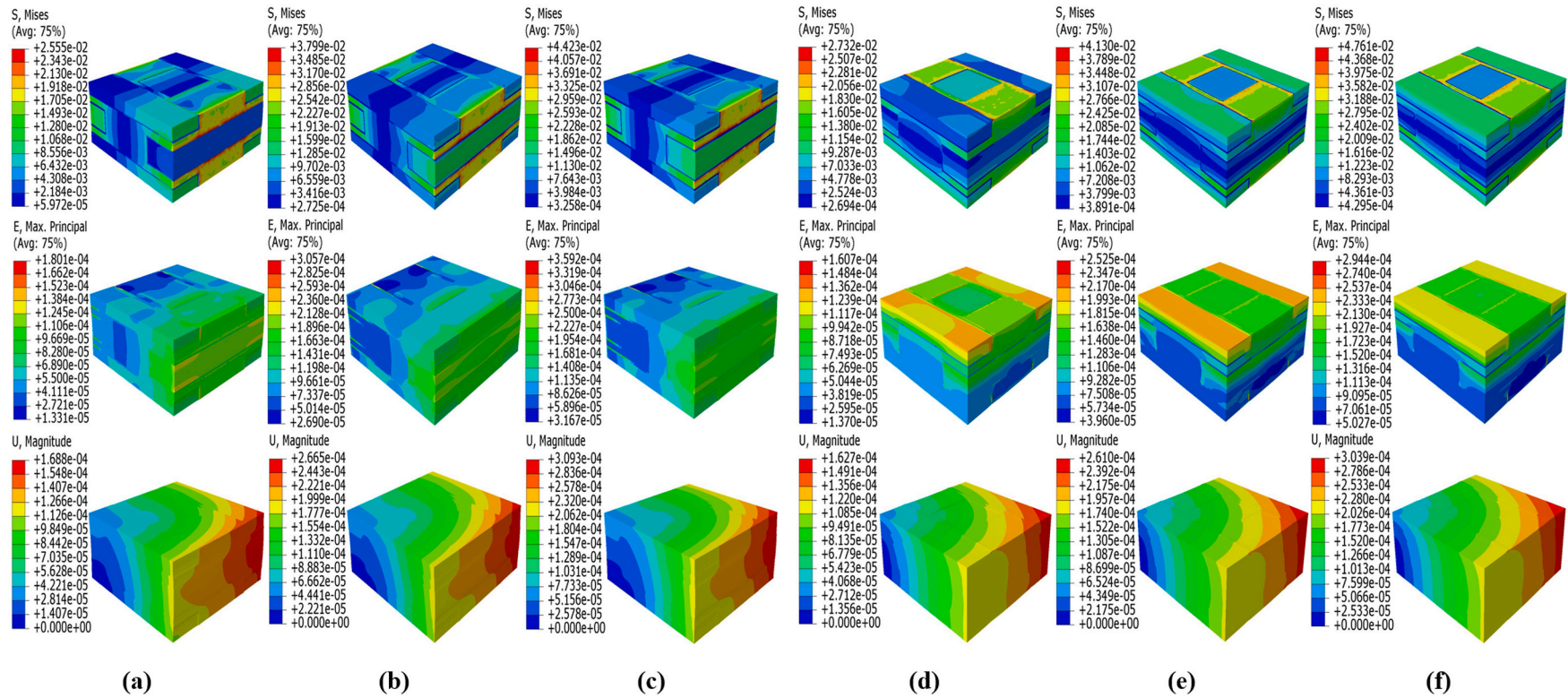


Fig. 13. Thermal mechanical contour of mesoscale model at temperatures of: (a)–(c) 100 °C, 600 °C, and 1200 °C, respectively, along the in-plane direction, (d)–(f) 100 °C, 600 °C, and 1200 °C, respectively, along the out-of-plane direction.

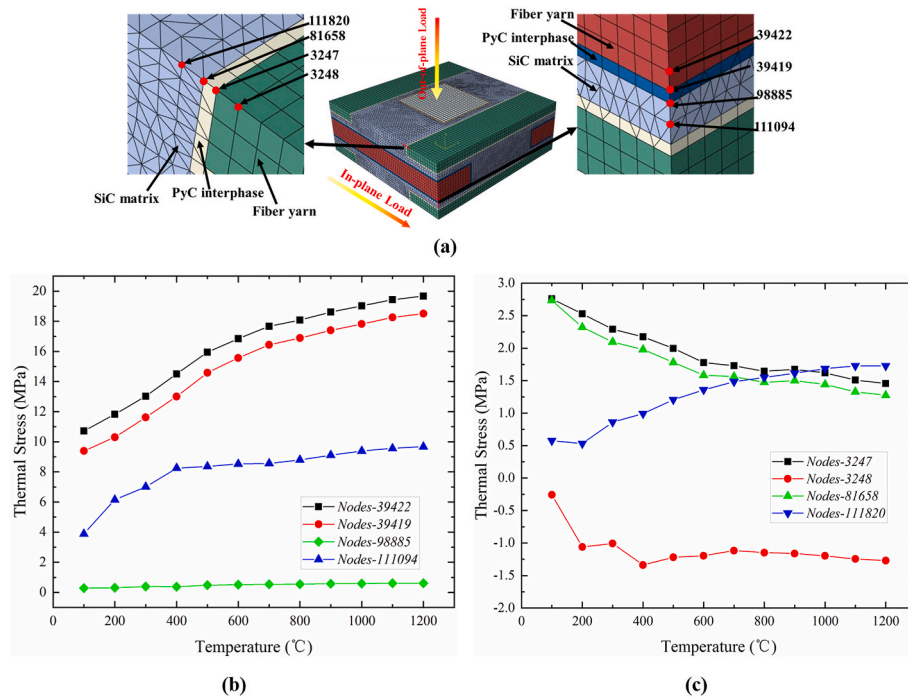


Fig. 14. Distribution of thermal stress between yarns, PyC interphase, void/matrix and their interface. (a) Location of each nodes, (b) In-Plane direction, (c) Out-of-plane direction.

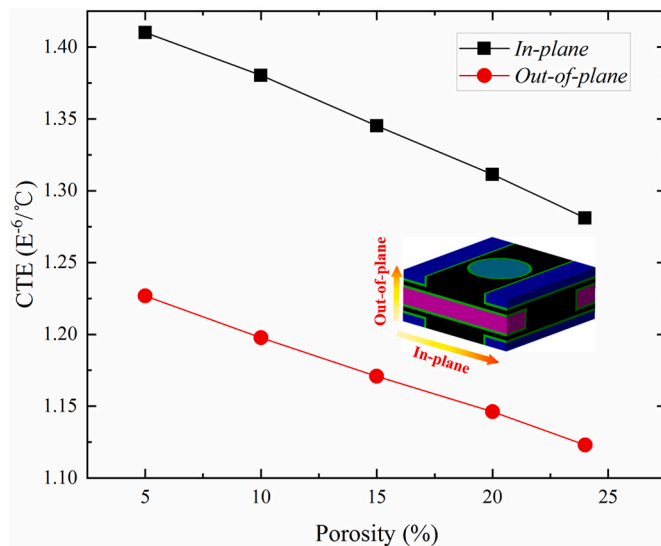


Fig. 15. Effect of porosity on the CTE.

temperatures higher than 200 °C. As shown in Table 2, the minimum errors are only 4.82% and 0.83% for in-plane and out-of-plane CTEs. The maximum error values of calculating the in-plane and out-of-plane CTEs are 18.64% (associated with a temperature of 800 °C) and 19.23% (associated with a temperature of 200 °C), respectively.

The thermal mechanical behaviors of the mesoscale model at different temperature points are shown in Fig. 13. As shown in the figure, the thermal strain, thermal stress, and thermal displacement along the in-plane and out-of-plane directions are all temperature dependent. Regardless of in-plane or out-of-plane thermal stress, the stress concentration occurs at the intersection of the SiC matrix and PyC interphase. The thermal stress in the matrix is larger than that in the interphase and yarns, owing to the larger CTEs of the matrix compared with those of the interphase and yarns. For the thermal strain, the

maximal thermal strain appears in the region of the yarns in parallel with the load directions. Similarly, both the in-plane and out-of-plane maximum thermal displacements increase with the growth of the temperature.

In order to characterize the thermal expansion effects of the C/SiC composites, the thermal stress along in-plane and out-of-plane directions are extracted from the intersection points of the yarns, PyC interphase, void/matrix. As is shown in Fig. 14(a), two sets of nodes along in-plane and out-of-plane directions, i.e. 39422, 39419, 98885, 111,094 and 3247, 3248, 81,658, 111,820 are extracted, respectively. For the load along the in-plane direction, the thermal stress of four points increases as temperature increasing. The thermal stress of nodes 39,422 is larger than that of nodes 39,419 and the thermal stress of nodes 98,885 is less than that of nodes 111,094. Therefore, the PyC interphase can effectively reduce the thermal stress of the intersection point, subject to the in-plane load conditions. For the load along the out-of-plane direction, the thermal stress located in the intersection point of yarns, PyC interphase and void/matrix (nodes: 3247 and 81,658) presents out the same changing trend with the growth of the temperature. The thermal stress in void/matrix (nodes 111,820) increases as temperature increasing, when the thermal stress in the yarns (nodes 3248) tends to decrease with the increment of the temperature.

Accordingly, there are some differences between the in-plane and out-of-plane loading. For instance, the thermal stress of the yarns are positive and the values are larger than that of the PyC interphase, void/matrix and their interface in the in-plane loading. However, in the out-of-plane loading the thermal stress of the yarns are negative and the values are less than that of the PyC interphase, void/matrix and their interface.

5.4. Effect of porosity

Five void/matrix models with cylinder-corner cubic are built to investigate the CTEs of the model with 5%, 10%, 15%, 20%, and 24% porosity at a temperature of 100 °C. The in-plane and out-of-plane CTEs of the 3D C/SiC composites with various porosities are presented in Fig. 15. As shown in the figure, both sets of CTEs decrease with

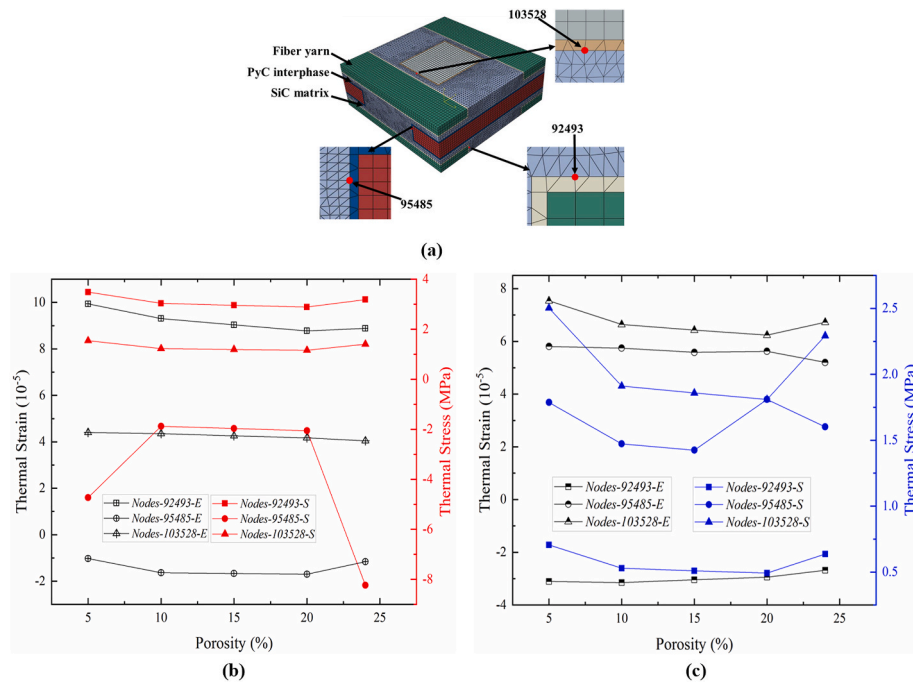


Fig. 16. Distribution of thermal stress between yarns, PyC interphase, void/matrix and their interface with different porosities. (a) Location of each nodes, (b) In-Plane direction, (c) Out-of-plane direction.

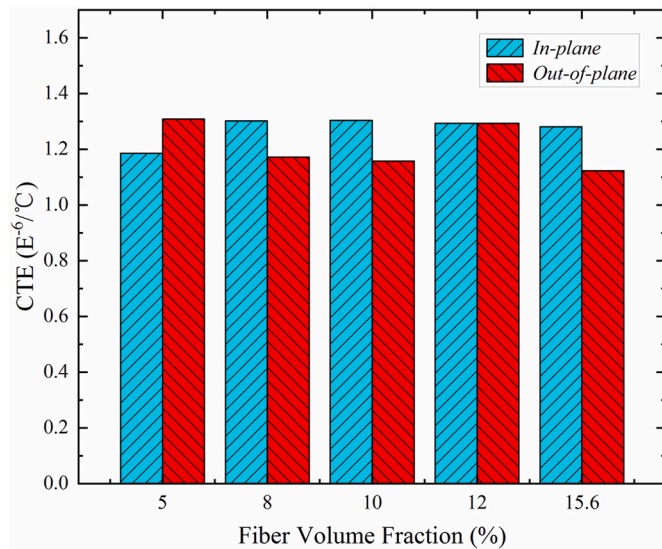


Fig. 17. Effect of fiber volume fraction on the CTE.

increasing porosity. The in-plane and out-of-plane CTEs drop by 7.11% and 8.36%, respectively, when the porosity increases from 5% to 24%. Therefore, the effect of porosity on the out-of-plane CTE is greater than that on the in-plane CTE.

For describing the variation of the thermal expansion behaviors under different porosities, the thermal strain and thermal stress along in-plane and out-of-plane directions are extracted from the intersection point between the void/matrix and the PyC interphase. As shown in Fig. 16 (a), three nodes, i.e. 92493, 95485 and 103528 are extracted, respectively. The thermal strain of three nodes remains almost unchanged with different porosities when applying the load along the in-plane and out-of-plane directions. Additionally, the thermal stress of nodes 95485 experiences a sharp rise and a rapid decline as porosities increasing from 5% to 24% on the condition of the in-plane load. The

thermal stresses of other two nodes are nearly unchanged under the same conditions. From Fig. 16 (c), it can be inferred that the thermal stress of nodes 95485 experiences a slight drop and a substantial growth with the increment of the porosities ranging from 5% to 24%. Consequently, the porosities have a significant effect on the thermal stress than thermal strain at the intersection points between the void/matrix and the PyC interphase.

5.5. Effect of fiber volume fraction

As mentioned in Section 2, we can use various fiber yarns to replace the guide sleeves, i.e., 1 K, 3 K, or 6 K fiber yarns. Therefore, a study considering the dependence of the CTEs on the fiber volume fraction along the out-of-plane direction is warranted. Four additional models with 5%, 8%, 10%, and 12% fiber volume fraction were built by changing the geometrical parameters of the out-of-plane fiber yarn cross-sections. The volume fraction along the out-of-plane direction of the origin meso-scale model is 15.6%. For a temperature of 100 °C, the predicted CTEs of the C/SiC composites with different fiber volume fractions are presented in Fig. 17. The results indicate that the in-plane and out-of-plane CTEs of the 3D C/SiC composites are almost equal when the out-of-plane fiber volume fraction is 12%. This suggests that equal in-plane and out-of-plane CTEs of these composites can be achieved by selecting appropriate fiber volume fractions for the different directions.

6. Conclusions

Multi-scale models based on periodic displacement and temperature boundary conditions are developed for predicting the CTEs of 3D C/SiC composites at temperatures ranging from 100 °C to 1200 °C. The results of the RVE models correspond closely to the results of the experiments. Furthermore, the effects of porosity and fiber volume fraction are also taken into consideration. Based on the FEM results and experimental results, the following conclusions are summarized:

- (1) Both the transverse and longitudinal CTEs of the cuboid and hexagonal RVE models are almost equivalent. The CTEs of the fiber random distribution RVE is different in comparison with the cuboid and hexagonal RVE models. The transverse CTEs slightly increases while the longitudinal CTEs substantially decreases as increasing the fiber volume fractions.
- (2) The in-plane and out-of-plane CTEs of the 3D C/SiC composites increase from 100 °C to 1200 °C from the numerical and experimental results. For a given temperature point, the in-plane CTEs are always larger than the out-of-plane CTEs.
- (3) The morphological characteristics of the voids have little influence on the CTEs of the void/matrix models. The voids are effective in lowering the CTEs of the 3D C/SiC composites. The in-plane and out-of-plane CTEs drop by 7.11% and 8.36%, respectively, when the porosities increase from 5% to 24%.
- (4) Equal in-plane and out-of-plane CTEs can be obtained by selecting appropriate fiber volume fractions for the different directions. Volume fractions of 15.6%, 15.6%, and 12% along the x, y, and z directions, respectively, yield almost equal in-plane and out-of-plane CTEs of the 3D C/SiC composites.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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